



Spark SQL

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Agenda

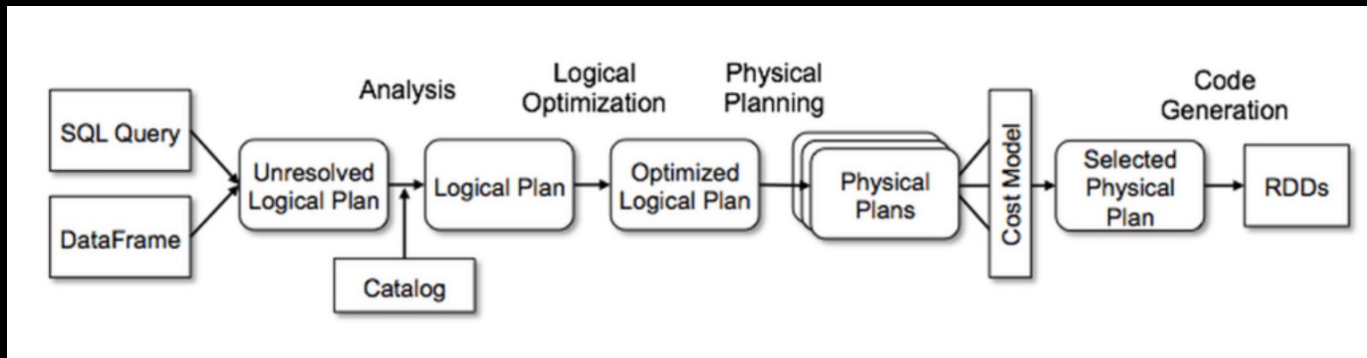
- Spark SQL Introduction
- DataFrame & Spark SQL
- Spark SQL Objects
- Read & Write Data

Spark SQL

- SQL interface available at Spark.
- Built using DataFrame API.
- Distributed Query Engine.
- Can be used to query File sources (Parquet, CSV, JSON, Avro), JDBC sources.
- `spark.sql("<SQL query>")`
- Uses ANSI compliant SQL (`spark.sql.ansi.enabled`).

Catalyst Optimizer

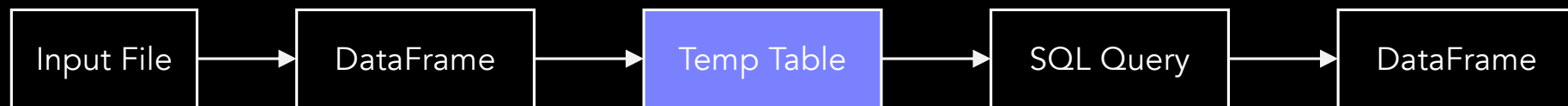
- Spark's Optimizer.
- Creates access path of DataFrame API and Spark SQL.



- Unresolved Logical Plan – Spark does not analyze the query/API for correct source names, its columns and data types.
- Logical Plan – It is created after analyzing the sources, syntax, data types, names etc.

SQL on DataFrames

- Create DataFrame and save them as temporary tables/views.
- Use SQL to query data.
- Result will be returned as DataFrame.



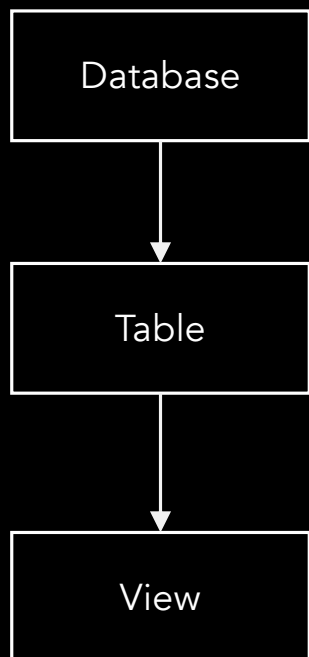
- Session temporary table – Available to only one Spark Session that created it.
 - `df.createOrReplaceTempView("<table>") – SELECT * FROM <table>`
- Global temporary table – Available to all Spark Sessions.
 - `df.createGlobalTempView("<table>") – SELECT * FROM global_temp.<table>`

DataFrame to SQL Table



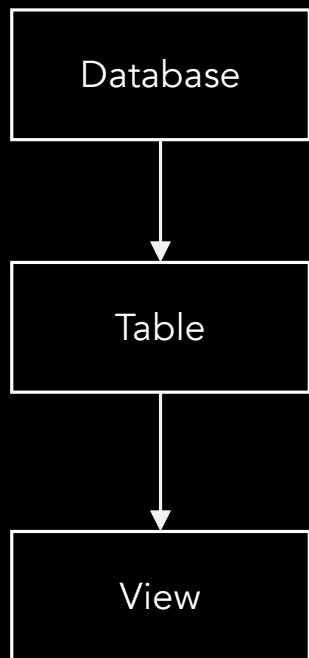
- saveAsTable – Store DataFrame as an SQL table. Part of DataFrame writer.
- Save metadata – Metadata repository. AWS Glue Catalog.
- Save data – Store data in distributed storage. spark.sql.warehouse.dir.
- Options:
 - option – (path, <location>)
 - partitionBy.
 - format – Parquet, ORC, Avro, JSON, CSV.
 - mode – append, overwrite, ignore, error.

Spark SQL Objects



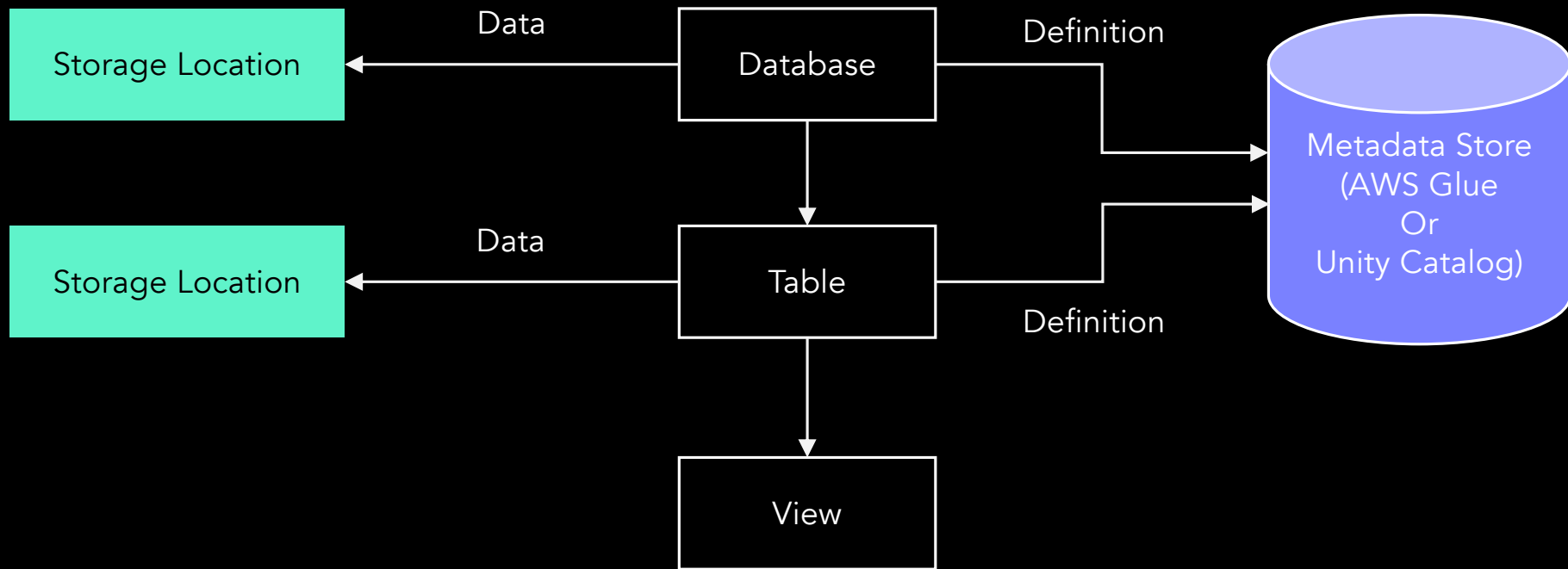
- CREATE DATABASE <dbname> LOCATION <storage-location-path>;
- DESCRIBE DATABASE EXTENDED <dbname>;
- ALTER DATABASE <dbname> SET LOCATION <storage-location-path>;
- Data Source table & Hive Format table.
- CREATE TABLE <tblname> USING CSV | ORC | JDBC | PARQUET | JSON
- PARTITIONED BY (<col1>, <col2>, ...)
- SORTED BY (<col1> ASC, <col2> DESC, ...)
- CLUSTERED BY (<col1>, <col2, ...>) INTO <n> BUCKETS
- LOCATION <storage-location-path>
- AS <select statement>

Spark SQL Objects



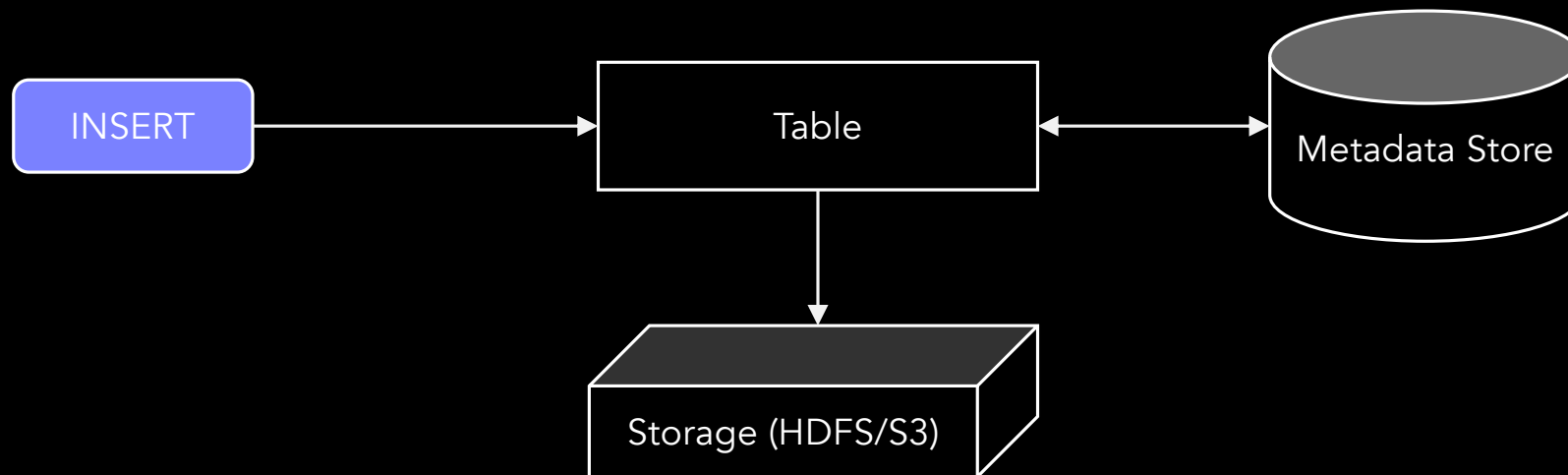
- CREATE TABLE <tblname> LIKE <existing-table>.
- USING parquet | csv ...
- STORED AS ORC | PARQUET ...
- LOCATION <storage-location-path>
- ALTER TABLE, DROP TABLE etc.
- CREATE VIEW <vw_name> AS <select statement>
- ALTER VIEW.
- DROP VIEW.

Spark SQL Objects



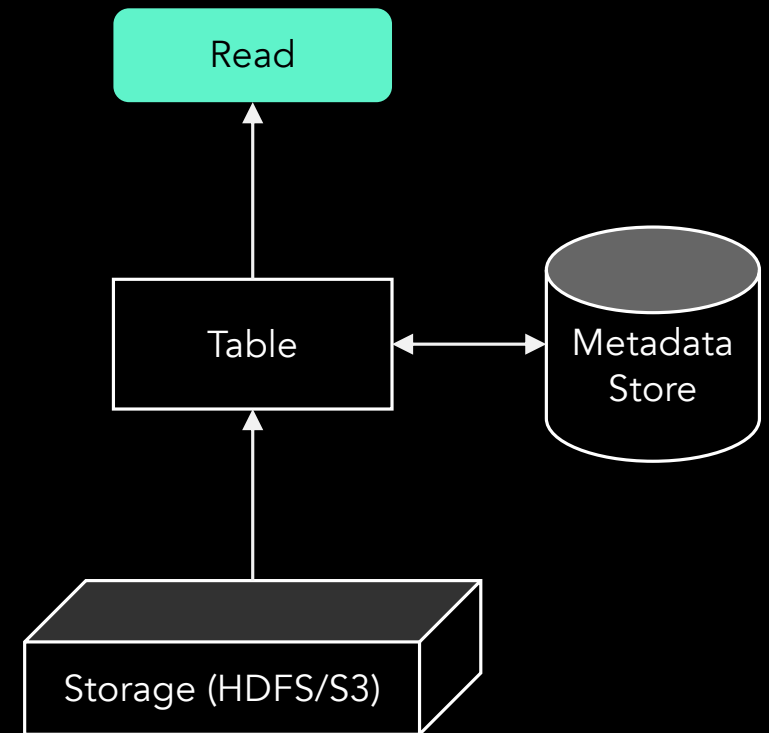
Write Data

- INSERT INTO <table> VALUES (<value1>, <value2>, <value3>, ...)
- LOAD DATA INPATH <path> OVERWRITE INTO TABLE <table-name>;
 - Only for Hive tables.



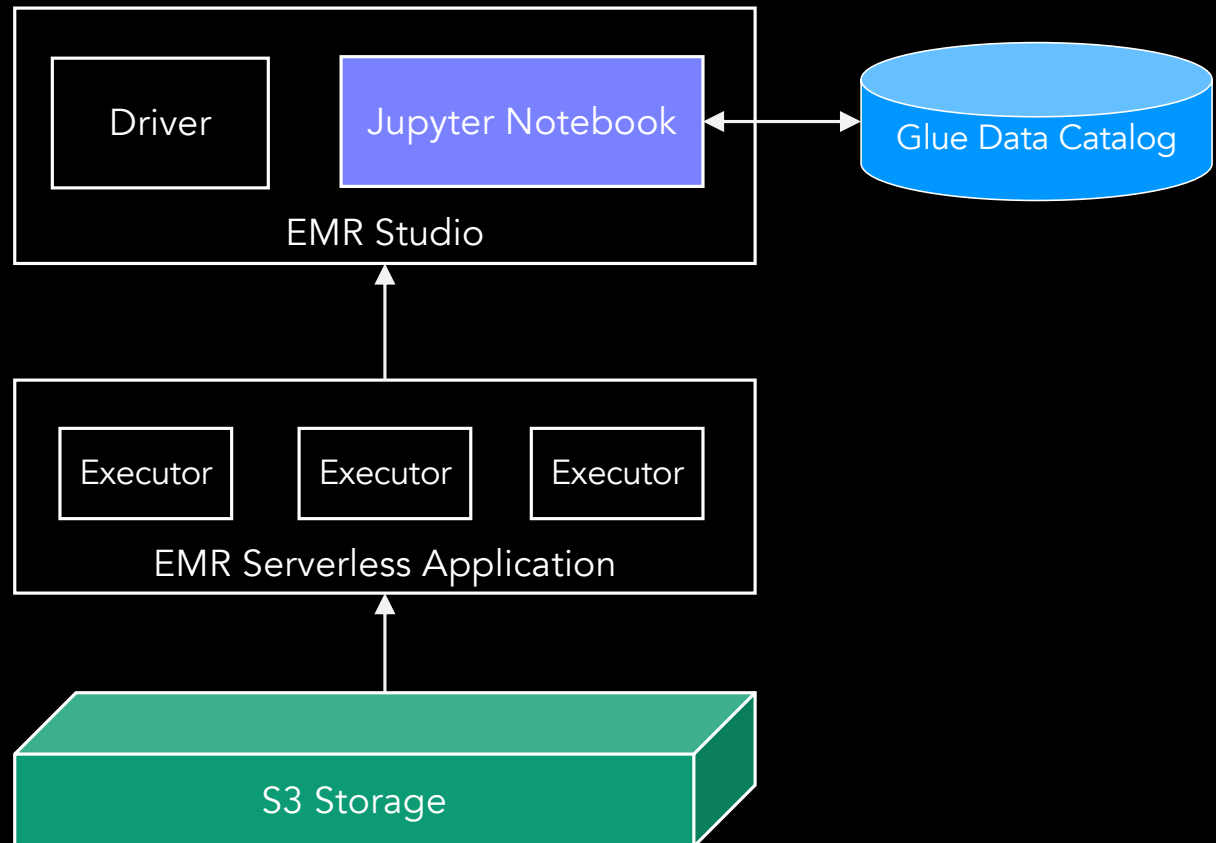
Read Data

- SELECT Statement
 - GROUP BY, HAVING, ORDER BY, SORT BY
 - WHERE, CASE
 - JOIN, LIMIT, LIKE,
 - CTE using WITH statement
- Window Functions
 - RANK, DENSE_RANK, PERCENT_RANK, ROW_NUMBER
 - MAX, MIN, COUNT, SUM, AVG
 - LEAD, LAG, FIRST_VALUE, LAST_VALUE
 - ROWS BETWEEN ...



Spark SQL on EMR

- Jupyter Notebook
- Glue Data Catalog
- S3
- EMR Serverless
- EMR Studio



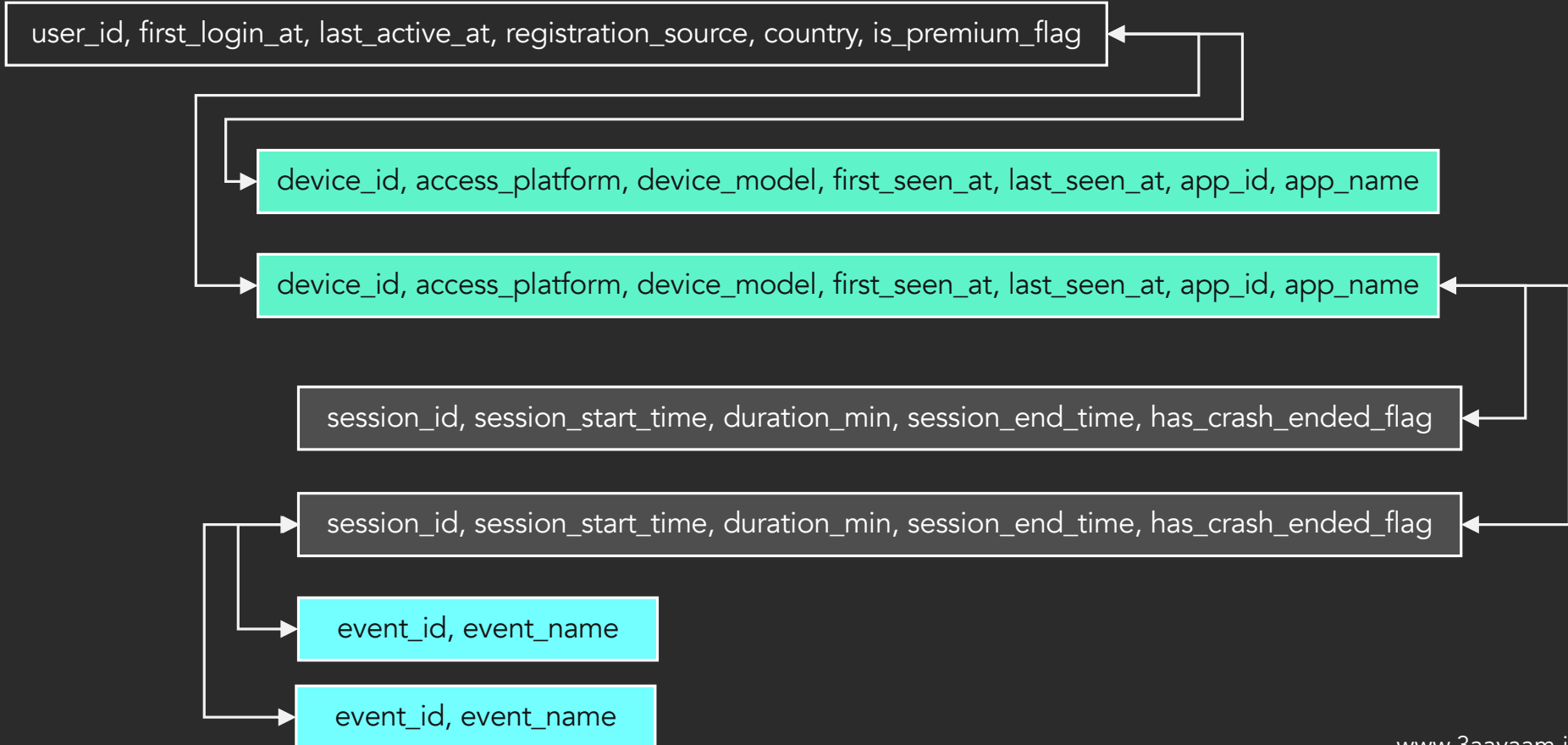
Spark SQL Exercise (Finance)

- Create the following tables using Parquet format. The input files are in CSV format. These tables should be in finance_db database. i) Account; ii) Customer; iii) Loan; iv) Transaction.
- Find all customer details that have taken Education loan.
- Top 3 customers for each loan type and their complete customer and loan details.
- List customers that have taken multiple types of loans. Find their total loan amount across all types and show their total account balance across all accounts for each customer.
- Find top 50 cross-country transactions and their details (like source customer details, target customer details and the cross-country transaction details.)
- Find top 10 cross-country transactions for each country.
- Top 10 oldest customer based on account opening date.
- Find all the running loans as per current date and how many monthly installments are remaining for these loans.

Project 7 – Mobile App Usage Tracking

- Dataset – 2 CSV Files. 50 GB (uncompressed).
- All columns of multiple entities are in one record.
- user_id
- first_login_at
- last_active_at
- registration_source
- country
- is_premium_flag
- device_id
- access_platform
- device_model
- first_seen_at
- last_seen_at
- app_id
- app_name
- session_id
- session_start_time
- duration_min
- session_end_time
- has_crash_ended_flag
- event_id
- event_name
- event_sequence_in_session
- crash_time

Project 7 – Data Structure



Project 7 – Use Spark SQL

- Load the single file into multiple tables as below and then solve the problem statements using these tables.
 - Users - `user_id` (PK), `first_login_at`, `last_active_at`, `registration_source`, `country`, `is_premium_flag`
 - Devices – `device_id` (PK), `user_id` (PK), `access_platform`, `device_model`, `first_seen_at`, `last_seen_at`, `app_id`, `app_name`
 - Sessions – `session_id` (PK), `device_id` (PK), `user_id` (PK), `session_start_time`, `duration_min`, `session_end_time`, `has_crash_ended_flag`.
 - Events – `event_id` (PK), `session_id` (PK), `device_id` (PK), `user_id` (PK), `event_name`, `event_sequence_in_session`, `crash_time`
- Identify specific app versions and device models that experience a significantly higher crash rate per session compared to the overall average, and investigate if these crashes are concentrated around certain event types.
- Analyze the impact of different `access_platform` and carrier combinations on session duration and crash frequency.

Project 7 – Use Spark SQL

- Analyze the crash frequency (total no. of crashes) for the top 3 selling access platform and top 3 selling device models per country. Place all columns for each country side-by-side for business and operational analysis.
- What is the top registration source for each country's top 5 apps.
- Which are the top 8 apps that have run smoothly longer than any other app for each access platform and which countries use those access platforms the most.
- How is the social media usage in each country compared to knowledge apps like Kindle, Adobe, Power Point and MS Word?
- Find out the apps for each platform that have crashed during opening, closing and in-between. Which tops the list at each operation?
- Which device model and platform combination is most stable in each country. What is the top registration source and premium user base for these countries?

Summary

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- DataFrame to Spark SQL Conversion
- Spark SQL Objects – Database, Table & View
- Write & Read Data

A scenic landscape photograph of a lake at sunset. The sky is a mix of purple, pink, and orange. A large, snow-capped mountain is visible in the background, reflected in the calm water of the lake. The foreground shows a rocky shore with green reeds and purple flowers. The text "Thank You" and the website "www.3aayaam.in" are overlaid on the image.

Thank You

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